

Invitation for Expression of Interest (EOI)

Tender Reference: DC/20■/11

Project Title: Northern Link Drainage and Sewerage Works

Issued by:

Drainage Services Department (DSD)

Government of Hong Kong Special Administrative Region (HKSARG)

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1. Project Background

The Northern Link railway project by the MTR connects Northern and Yuen Long Districts, while the drainage and sewerage works are crucial infrastructures supporting the railway development, particularly around Kwu Tung, San Tin, and Kam Sheung Road. These works ensure effective management of stormwater and wastewater, supporting the operation of future MTR stations and urban development in the Northern Metropolis.

This tender call aims to invite qualified contractors in the development of the drainage and sewerage works. It provides detailed description of the project scope and requirements, instructions to the applicants, and assessment of the applications with a scoring sheet.

2. Introduction of the Organization

The Drainage Services Department (DSD) is responsible for planning, designing, constructing, and maintaining Hong Kong's drainage and sewerage systems. DSD is committed to sustainable development, flood prevention, and environmental protection.

3. Project Objectives

The objective of this tender focuses on construction of drainage and sewerage works along the stations of the Northern Link railway. The project integrates with existing and planned railway and housing developments. DSD aims to provide modern sewerage infrastructure to support urban expansion and ensure compliance with environmental and public health standards.

4. Project Scope

The project involves construction of drainage and sewerage works and shall include but not limited to:

a. Installation of gravity and rising mains, manholes: The construction of the drainage and sewerage infrastructures along the stations of the Northern Link, notably Kwu Tung and San Tin, and preparation for future Northern Metropolis development. Its location along the Northern Link functions as the core of future pipeline branches. The pipeline network focuses on maintainability and scalability, to adapt to the potential rise in usage and increasingly complex network.

b. Construction of pumping stations: The pumping stations power the rising mains which transport the water upstream, redirecting the water to the water treatment plants along with the unpressurized gravity mains. The functionality will be expanded for the future Northern Metropolis development; therefore it is crucial to ensure its scalability, Contractor is expected to maintain the pumping stations in reliable condition and operation.

c. Surveying: Topographic, building, and utility surveys. Topographic survey for enabling works, 3D LiDAR scanning for as-built records. Other than traditional surveying, DSD also requires submission of as-built records in 3D point cloud data, using 3D LiDAR scanning to ensure precise and detailed documentation records for future underground utility management and development.

d: Hydraulic modeling and sewerage system design: Detailed drainage and sewerage pipeline network designs are required, including storm and sewerage simulation software to ensure the drainage system can handle the exceptional storm events and the population growth in the future Northern Metropolis.

e. Construction Management and Environmental Mitigation: All construction activities are required to adhere to the guidelines under the EPD. The Contractor is expected to manage all environmental protection measures including but not limited to construction waste, noise, and dust control.

5. Project Deliverables

a. Preliminary and detailed design documents: Submissions shall include design drawings with invert levels of pipes and manholes, engineering calculations including cut and fill costs and transportation plans, technical specifications, legal competence, and compliance statements. the Contractor shall first gain approval by DSD to further progress from preliminary design to detailed design. Design documents shall adhere to all technical standards under DSD, environmental guidelines specified by the Environmental Protection Department (EPD), and safety regulations under Labour Department.

b. Construction drawings and specifications: Detailed construction drawings shall include but not limited to plans, cross sections, utility coordination drawings. Specifications of the drawings shall adhere to the DSD Standard Drawings.

c. Utility Infrastructure Installations: Drainage and sewerage infrastructures include but not limited to gravity and rising mains, manholes, and pumping stations. Each component shall be delivered and certified in accordance with project requirements and DSD specifications.

d. 3D LiDAR Scanning for As-Built Records: Point cloud data collected from the 3D LiDAR scanners can provide high-resolution 3D models of the open-cut pipes and manholes. These records will ensure the infrastructure installed adhere the DSD LiDAR scanning specifications and support future utility developments.

e. Survey reports: Submission of detailed survey reports include but not limited to topographic survey, building, utility for stages before, during, and after as-built. The records ensure the pipe alignment and the invert levels for both open-cut and trenchless are within the tolerances of the design.

The Contractor is expected to complete the project and proceed to final handover in 24 months after receiving the tender award. The Contractor shall provide a detailed project schedule in the tender submission and submit progress reports regularly throughout the duration of the contract.

6. Contractor Requirements/Qualifications

To ensure the drainage and sewerage works along the Northern Link can be successfully delivered, the Contractor are required to have qualified personnel, relevant experience in utility and engineering infrastructures, especially previous DSD contracts, and technical

capabilities with skilled and semi-skilled personnel. The following are the minimum requires and qualifications for the Contractor for this contract.

- Experience in drainage and sewerage infrastructure projects:

The Contractor shall have a minimum of 5 years of documented experience in the construction of drainage and sewerage projects, preferably contractors of previous drainage or sewerage projects. Experience shall include installations of gravity and rising mains, pumping stations, and associated utility works. Contractors of previous governmental department contracts such as DSD or WSD are highly desirable.

- Professional engineering qualifications:

The Contractor shall have acquired key personnel qualified for civil engineering, including but not limited to project managers, design engineers, foremen, safety managers, and surveyors. It is required for the Contractor to submit the engineering and surveying documents to the DSD, by members of Hong Kong Institution of Engineers and Hong Kong Institute of Surveyors.

- Specialized certifications for key personnel:

The Contractor shall have acquired holders of certifications relevant to specialized construction activities including but not limited to welding certification, confined space entry certification, and excavator operator license. The Contractor must demonstrate sufficient relevant workforce and compliance to the safety regulations under the Labour Department.

7. Scoring Sheet Development

The submitted proposals by the applicants will be evaluated through a scoring sheet various criterion. The criteria are weighted to reflect the importance to the project and the priorities of DSD. The details are as follows:

- Technical competence (40%): DSD assesses the technical ability of the Contractor including but not limited to the professional knowledge in drainage and sewerage engineering, technical compliance in the proposed design, timeline management, addressing the difficulties of the project, waste management. The Contractor shall be capable to use advanced survey equipment for accuracy monitoring and as-built records.
- Relevant experience (30%): Past performance on similar projects will be accounted for, especially large-scale drainage and sewerage infrastructure under governmental contracts.

The Contractor shall demonstrate successful completion in past contracts, compliance with the safety regulations and environmental mitigations.

- Resource allocation and team structure (20%): DSD assesses the adequacy and qualifications of the project team, notably the specialized personnel with certifications or license, and the access to the specialized equipment for construction and surveying. The Contractor shall have an organized structure of personnel allocation and management, with clear roles, qualifications, and responsibilities. The realism of the proposed timeline and the management of project progress will also be assessed.

- Cost effectiveness (10%): Applicants shall demonstrate clarity and realism in the budget and timeline estimations. DSD values both work completion and the proposed price. The Contractor is expected to complete the project in 24 months.

8. Instruction to the applicant organization

Each interested applicant shall submit exactly one application representing the company, which contains an Expression of Interest (EOI) focused on developing the drainage and sewerage works of the Northern Link. Applicants shall follow the guidelines below for the EOI:

- Specific Criteria: Make sure that your organization has the legally required qualifications as mentioned in the EOI. This includes but not limited to safety compliance to the Labour Department, environmental mitigation, personnel qualifications.

- Content requirements:

- Company profile: Provide a detailed overview of the company, organizational structure, scope of business.

- Relevant experience: Describe previous projects related to drainage and sewerage infrastructure, especially contracts related to utility and civil engineering.

- Personnel Qualifications: Provide detailed certifications and licenses of the key personnel, including engineers, safety managers, and specialized technicians.

- Timeline and budget management: Provide the detailed management over the project progression and the estimated bidding cost, including resource allocation over labour, material, equipment.

9. Challenges and Solution

The development of the Northern Link drainage and sewerage works can impose several technical and environmental challenges. Applicants must clearly address these issues in the proposal and provide solutions.

Key challenges include:

- Integration with concurrent infrastructure projects: The development of the Northern Link will be parallel with its drainage and sewerage development, the Contractor is expected to coordinate with the MTR and its subcontractors to ensure minimal project disruption.
- Environmental impacts in protected areas: The development of the Northern Link is close to the farmlands and the wetland parks. Applicants shall address this issue with precautions developing in the area and include the risk assessments and mitigation strategies in the proposal.

10. Enquiries

For any inquiries concerning this EOI invitation, please contact:

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